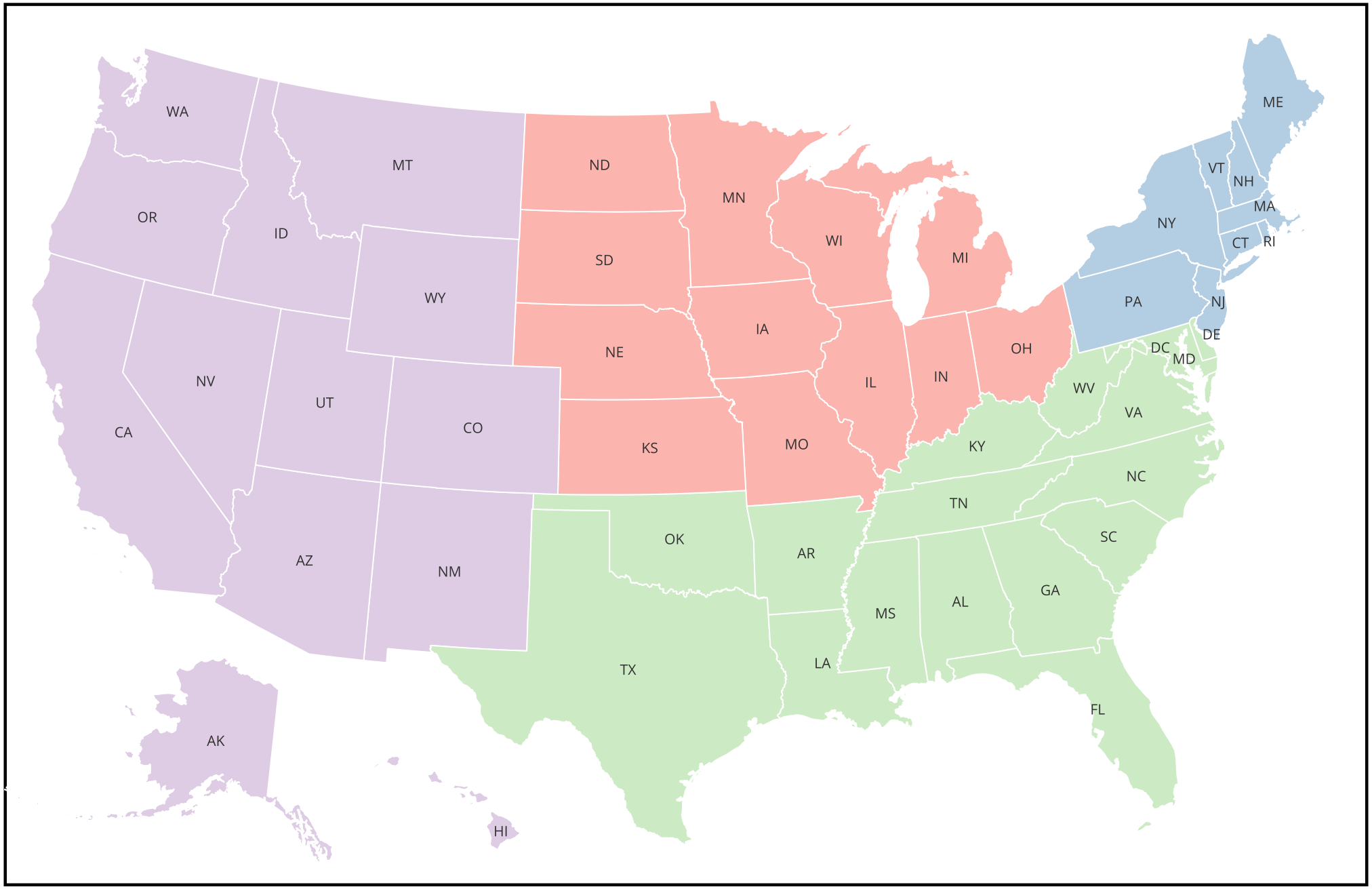
U.S. States Categorized by Region

Cartography by Sophia Santos

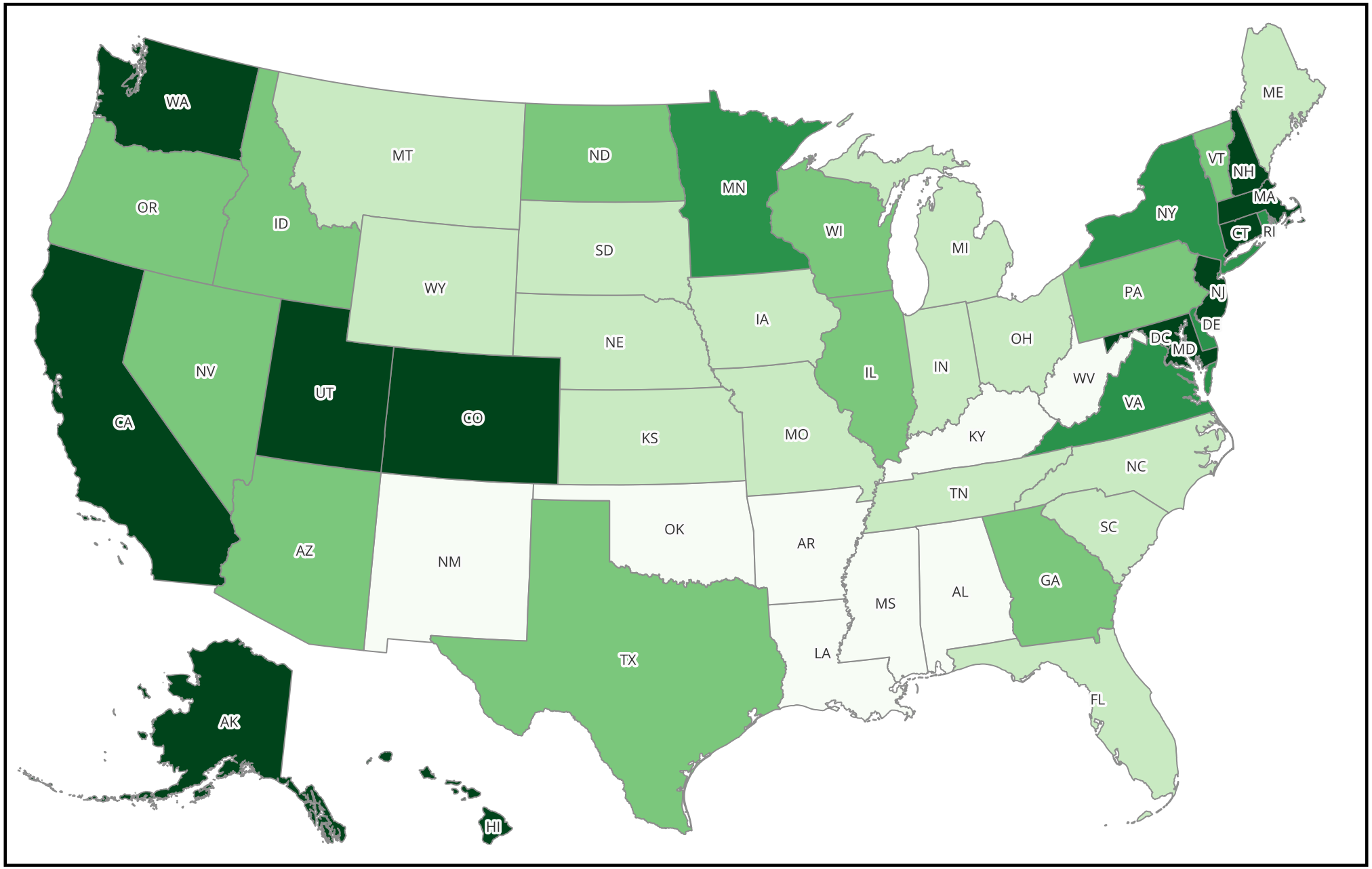


Midwest Northeast South West

Data Source: Introduction to GIS course website, derived from U.S. Census Bureau

U.S. States Classified by Median Household Income

Cartography by Sophia Santos

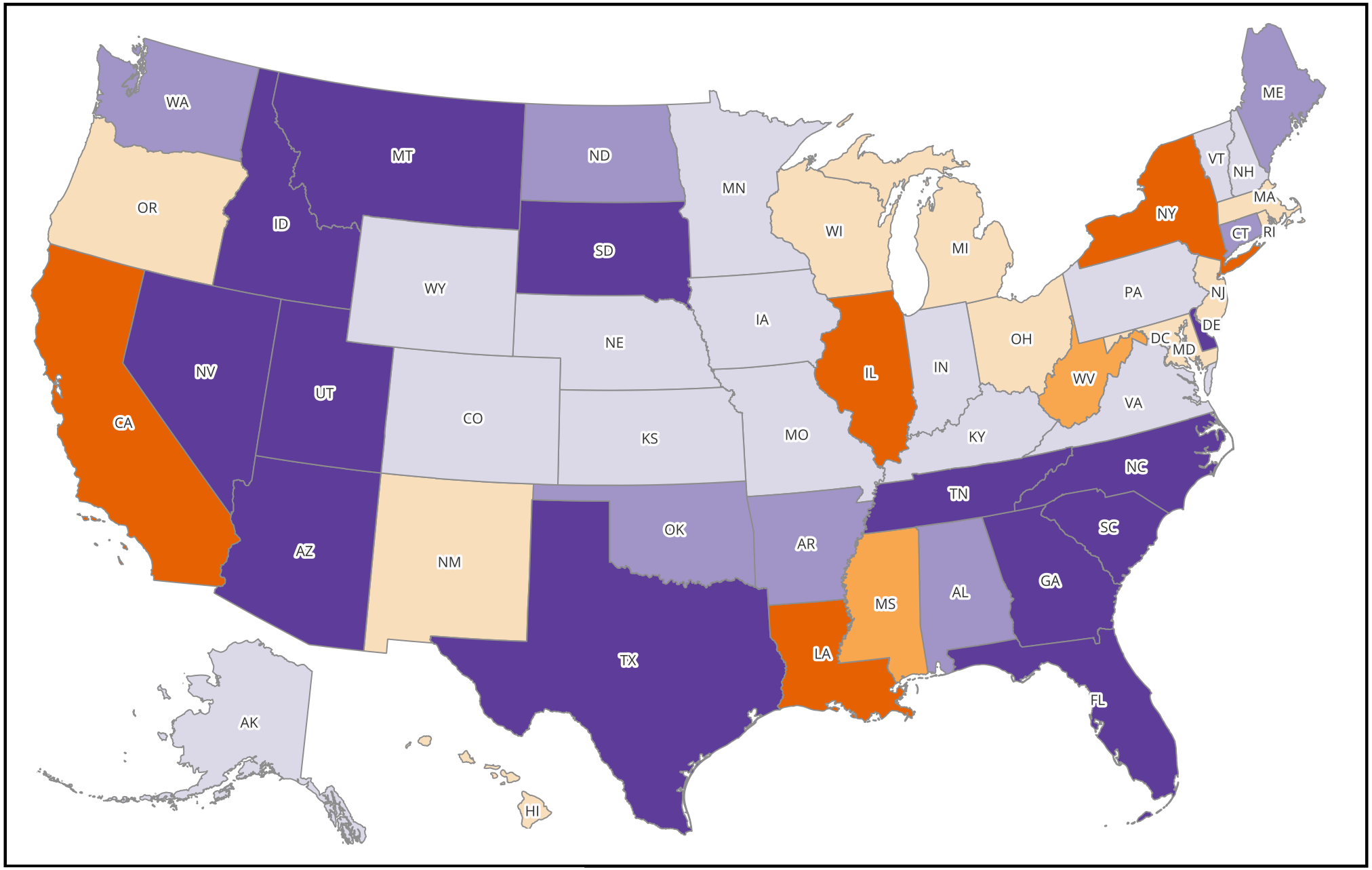


$52,719 - $59,726 | $59,726 - $70,042 | $70,042 - $76,708 | $76,708 - $85,873 | $85,873 - $101,027

Data Source: Introduction to GIS course website, derived from U.S. Census Bureau

U.S. States Classified by Change in Population from 2021 to 2022

Cartography by Sophia Santos



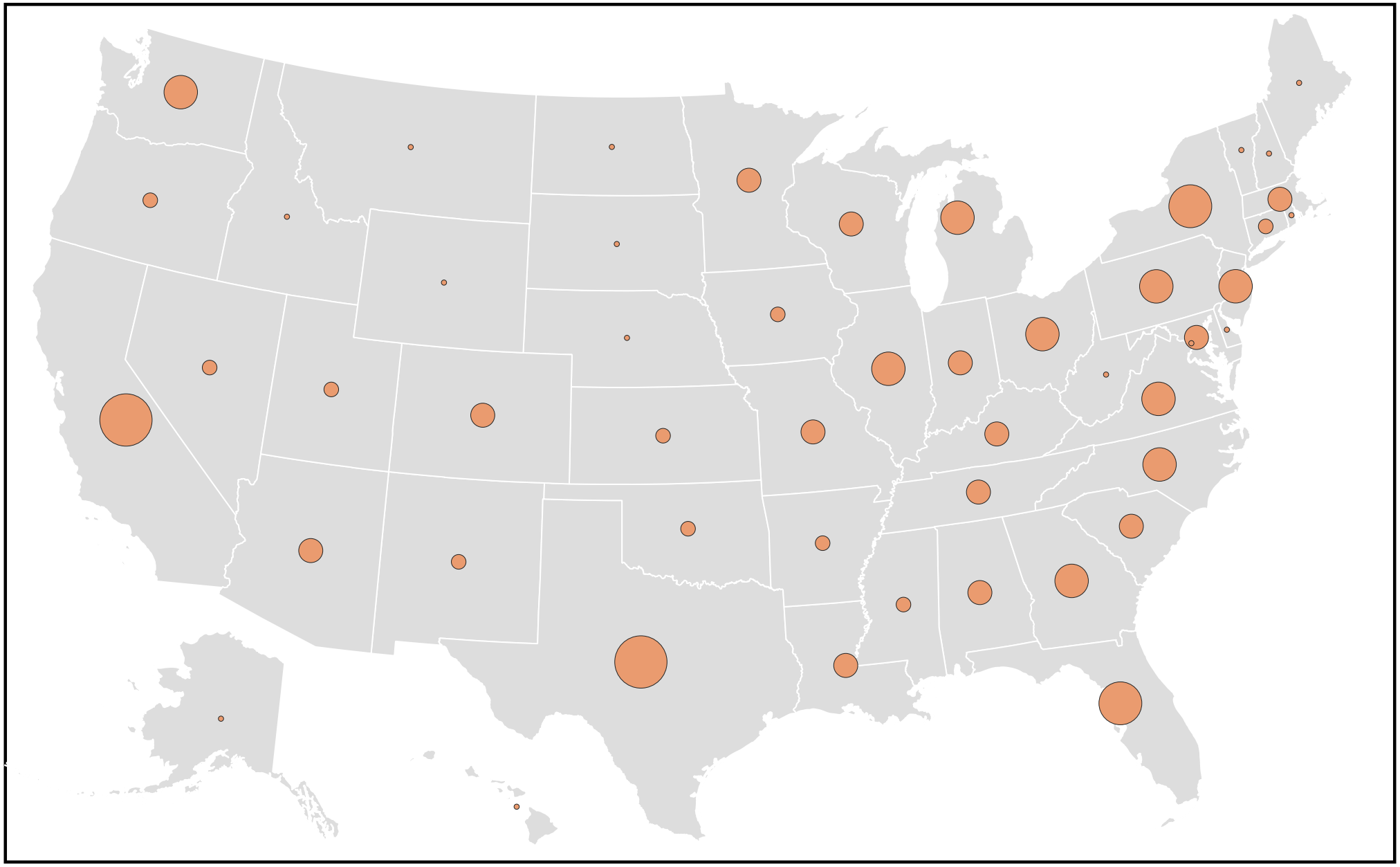
-1% to -0.5% | -0.5% to -0.25% | -0.25% to 0%

0% to 0.5% | 0.5% to 1% | 1% to 2.5%

Data Source: Introduction to GIS course website, derived from U.S. Census Bureau

U.S. States Classified by Total Population Size in 2022

Cartography by Sophia Santos



580,000 - 2,000,000

2,000,000 - 4,500,000

4,500,000 - 7,500,000

7,500,000 - 13,000,000

13,000,000 - 30,000,000

30,000,000 - 40,000,000

Data Source: Introduction to GIS course website, derived from U.S. Census Bureau

Sophia Santos

GEOG 7: Assignment 5 Write-Up

The visualized data for Map #2 comes from the “MHI” variable in the U.S. Census Bureau's American Community Survey (ACS) dataset, which portrays the estimated median household income within each state. The data is naturally distributed with natural breaks and clusters. So, I followed the natural structure of the data and used the Natural Breaks (Jenks) method of classification. I chose to classify the data into 5 groups, which effectively utilizes the gaps in the data and properly breaks up large clusters.

Map #3 visualizes the “POP_PCTCHG_2122” variable in the same dataset, which gives information about the percent change in estimated state population from 2021 to 2022. While the data does have its own natural breaks, the gaps are small. Thus, the difference from one cluster to the adjacent cluster is often small. Additionally, since this variable represents percentage values, I found it best to create a custom classification scheme based on standard whole, half, and quarter percentages. Therefore, I utilized equal count (quantile) classification to first split the data into 6 roughly even sized groups. From there, I made small adjustments to the percentage intervals, making them more intuitive.

In Map #4, we visualize”POP_2022,” a variable of the estimated state populations in 2022. This data is heavily skewed to the right. Because of this, a Quantile or Equal Interval classification would not suit the non-uniform distribution. I first tried the Logarithmic Scale method, but QGIS would not allow me to pick a number between 4 and 10 classes. Since I was aiming for 6 to 8 classes, I decided that this restriction would not suit my mapping. I decided to instead use the Natural Breaks (Jenks) method of classification which effectively splits up the large left cluster of the data and makes use of the breaks in the sparse right side.